

Market Decision Structure

Fresh report for the 918-tick run. This supersedes the earlier 101-tick pilot memo.

Research anchors: MARKET_MANIFOLD_RESEARCH_PLAN.md and MARKET_MANIFOLD_IMPLEMENTATION_PLAN.md

March 20, 2026

48 layers

Held-out grouped evaluation

Decision-structure slice

This report addresses one narrower question from the market-manifold program: **when does the model bind an action-relevant preference to a specific asset?** The new 918-tick decision-structure run points to a mixed answer. Buy-side preference remains strongly early, but target-asset and sell-target signals both receive modest downstream sharpening from later policy and affordance context.

Executive Read

Ticks

918

559 trades and 359 observations.

Asset rows

5,719

Row-level pooled states across the held-out evaluation.

Best pre AUROC

0.878

is_buy_target from row_eos at layer 25.

Largest post lift

+0.019

Target-asset and sell-target both improve modestly post-context.

Main read: the architecture does not look like “all preference early” or “all policy late.” It looks staged. Market rows already carry a strong asset preference signal, but downstream sections such as constraints_eos and active_settings_eos can sharpen that signal when legality or actionability matters.

Dataset

This is the first decision-structure run that is large enough to move beyond the pilot.

Decision mix

559 trades vs 359 observations.

Trade direction

280 buys vs 279 sells.

Positive rows

558 target-asset rows, 279 buy-target rows, 279 sell-target rows.

Why this matters:

- the run is no longer bottlenecked by sell scarcity
- the bearish-side readout is now meaningful
- the result is a real decision-structure test rather than a buy-heavy pilot

Current sample composition

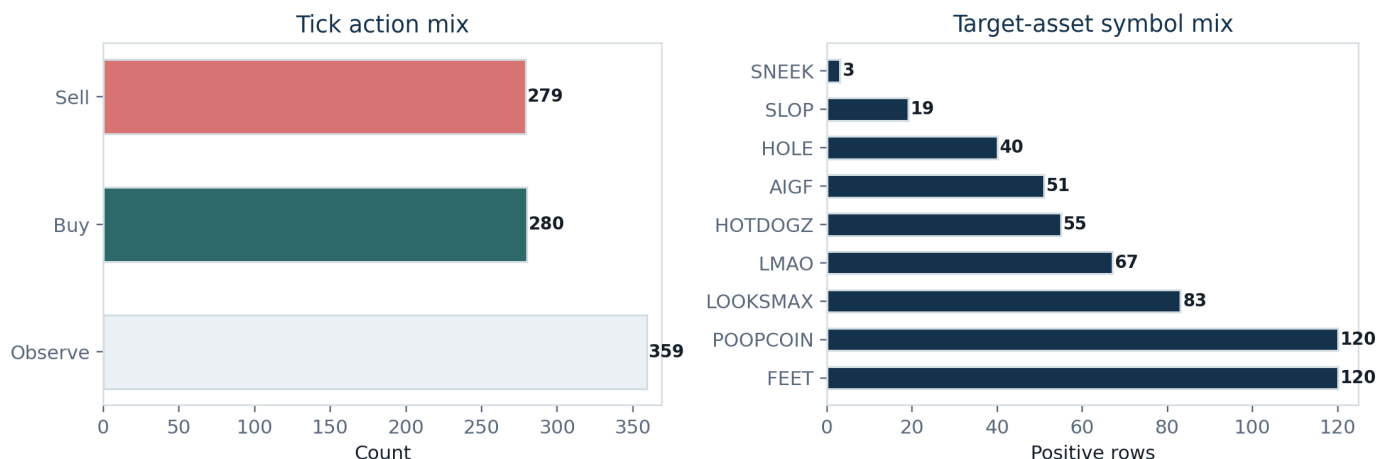


Figure 1: Dataset composition for the 918-tick run. The class balance is much healthier than in the pilot, though target-asset rows still concentrate in a handful of symbols.

Method

The report focuses on the decision-structure slice of the broader research plan. The analysis compares what is decodable from market rows alone against what becomes decodable after downstream sections are added.

01

Pool the sequence

Reduce full-sequence captures into `row_mean_i`, `row_eos_i`, and downstream section states such as `active_settings_eos`, `portfolio_eos`, `constraints_eos`, `prev_decisions_eos`, and `last_token`.

02

Train held-out probes

Fit linear probes on grouped splits so multiple rows from the same tick do not leak across train and test.

03

Compare pre vs post

Treat row states as pre-context and row-plus-section combinations as post-context representations.

04

Read staging

If post states beat pre states, downstream context adds usable structure; if pre states dominate, the preference is already formed while the market is read.

Main Quantitative Findings

Target	Best pre state	AUROC	Best post state	AUROC	Post - Pre
is_target_asset	row_mean @ layer 35	0.815	row_mean + constraints_eos @ layer 26	0.834	+0.019
is_buy_target	row_eos @ layer 25	0.878	row_mean + constraints_eos @ layer 29	0.870	-0.008
is_sell_target	row_mean @ layer 2	0.839	row_mean + active_settings_eos @ layer 2	0.857	+0.018

Interpretation of the table:

- `is_buy_target` remains strongest in a pre-settings row state
- `is_target_asset` improves slightly once constraints are added
- `is_sell_target` improves slightly once active settings are added

This is the key reason the new report differs from the pilot. The earlier “everything is early” story no longer holds as a blanket conclusion.

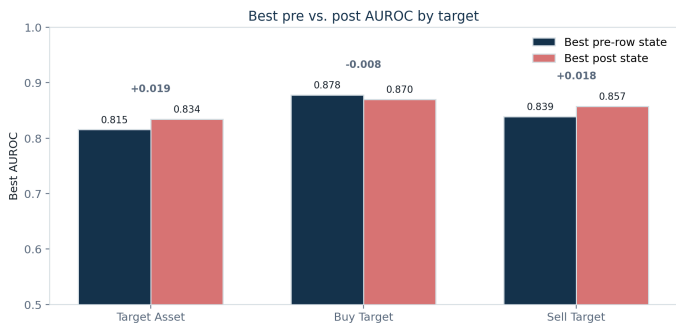


Figure 2: Best pre-row vs best post-context AUROC by target. Post-context wins are small rather than dramatic, but they are now real for target-asset and sell-target.

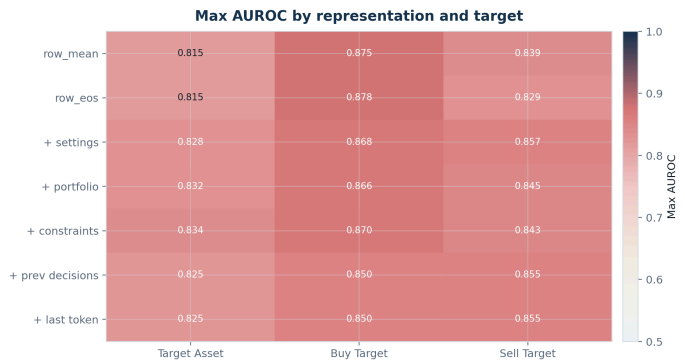


Figure 3: Maximum AUROC by representation and target. row_mean and row_eos remain central, but + constraints and + settings now matter for some tasks.

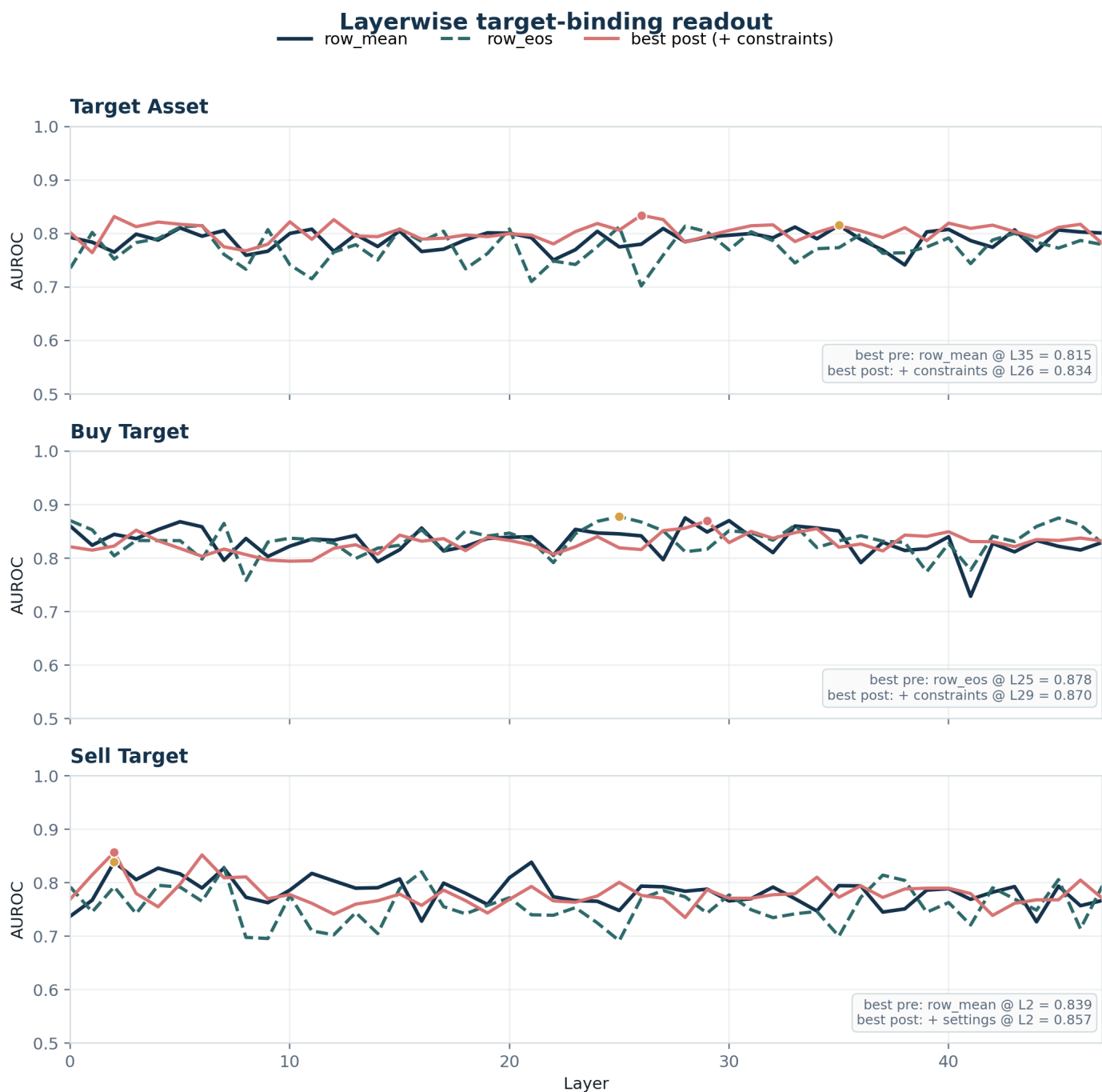


Figure 4: Layerwise AUROC curves. The buy-target signal remains distinctly early, while target-asset and sell-target gain from downstream context without requiring a deep late reconstruction.

Interpretation

Supported now

Early asset preference is real. The model does not need to wait for the end of the prompt to identify an attractive buy candidate.

Also supported now

Downstream context is not cosmetic. Constraints and settings can sharpen the decision signal for some targets.

Best current interpretation:

- the model forms a strong early market preference while reading asset rows
- later sections reweight that preference when legality, sizing, or actionability matters
- the result is consistent with the research plan's split between the asset-in-context manifold, the asset-valence manifold, and the final decision manifold

Best current interpretation: Xenon appears to use an early market preference layer followed by a later policy-and-affordance refinement layer. That is closer to “early read, later reweighting” than to either extreme of “all preference at the start” or “all policy at the end.”

What We Accomplished

Relative to the root plans, this run closes the first serious decision-structure milestone.

- balanced cohort selection and large-scale full-sequence capture on Modal
- reliable pooled row and downstream section states
- held-out pre/post probe analysis for `is_target_asset`, `is_buy_target`, and `is_sell_target`
- a result strong enough to constrain the next research step instead of merely validating pipeline health

What Is Still Missing

This report does **not** finish the broader market-manifold program.

- blocked-bullish and blocked-bearish labels for observe cases
- base market-geometry work: intrinsic dimension, RSA, pairwise probes, and router specialization
- stronger settings interventions for a clean pre/post reinterpretation test
- causal necessity tests such as row masking and rank-vs-magnitude corruption

Recommended Next Steps

A

Build blocked-valence labels

Use constrained observe examples to separate latent bullishness or bearishness from mere non-action. This is the missing bridge from executed action to the actual asset-valence manifold.

B

Run market geometry in parallel

Start intrinsic dimension, RSA against raw and rank spaces, and router specialization on current row activations.

C

Design stronger settings interventions

Replace weak prompt edits with real policy changes so pre/post reinterpretation is tested against meaningful shifts rather than near-identity prompts.

D

Then run causal tests

Once the label program is stronger, use row masking, pairwise swaps, and rank-preserving vs magnitude-preserving corruptions to measure what the model actually needs.

The main value of this run is that it narrows the search space. We no longer need to argue about whether the decision-structure pipeline works. It does. The next question is how to move from executed action labels to true asset-valence labels, while building the base market-geometry measurements in parallel.